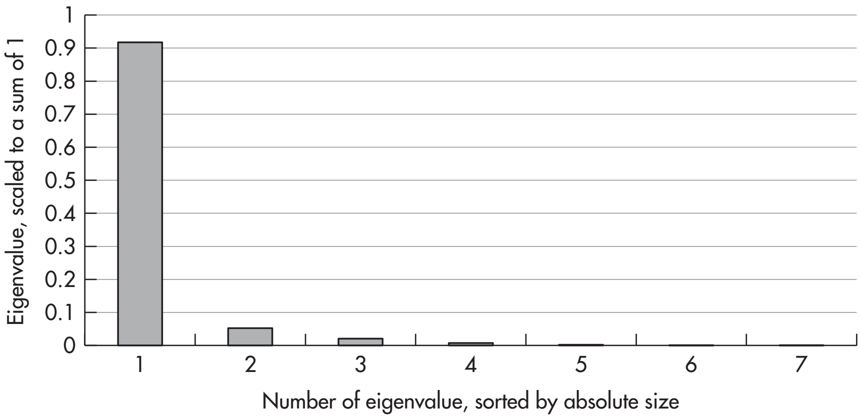

FIGURE 13.5 Scaled eigenvalues of a PCA on CDS quotes for EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 4 Mar 2009 to 26 Sep 2012, weekly data.

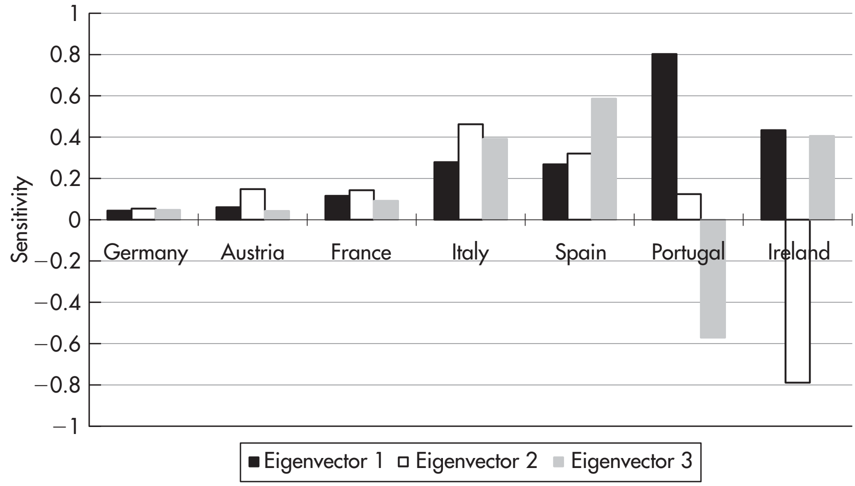

FIGURE 13.6 First three eigenvectors of a PCA on CDS quotes for EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 4 Mar 2009 to 26 Sep 2012, weekly data.

The qualitative results are unsurprising, with Germany being less affected than Austria, Austria less than France etc. Figure 13.6 offers the additional insight into the quantitative relationships. An overall worsening of the crisis that leads to a 10 bp (basis point) widening of German CDS usually leads to a 26 bp widening of French CDS, a 64 bp widening of Italian CDS and so on, with the relationships being given by the quotients of the sensitivities to the first factor. It is interesting to note that the inclusion of non-European countries in the PCA does not change the results significantly; in particular, no "euro-block" is evident (i.e. there is no eigenvector grouping together all euro countries through positive sensitivities versus all non-euro countries through negative sensitivities). Thus, euro sovereigns seem to be priced in the CDS market as individual countries, not as part of a euro-block.

The higher eigenvectors have positive entries for all countries except one. Thus factors 2 and 3 are country-specific factors, measuring the differentiation of a particular country's CDS versus the overall CDS level given by factor 1. As it happens, a country has a specific factor (with some meaningful eigenvalue), if and only if it is a bailout country (Portugal and Ireland). Hence, the PCA decomposes the EUR sovereign CDS market into its general level, subject to the overall worsening and improvement of the crisis (factor 1), and into the pricing action subject to the bailouts of specific countries (factors 2 and 3). Using this framework, a particular move in the CDS market can thus be attributed to the different pricing mechanisms as expressed by the eigenvectors. For example, a narrowing of Irish CDS quotes can be decomposed into the part due to a general improvement of the euro crisis (factor 1) and into the part due to Ireland-specific developments (factor 2).

While this is a reasonable result, it also means that "pure" relative value trading on the whole EUR CDS universe is hardly possible. Factor 1 represents the overall crisis and factors 2 and 3 country-specific developments, both of which are highly influenced by political decisions. For example, European Central Bank support drives factor 1 lower, while bailout programs in Ireland cause factor 2 to increase, and civil unrest in Portugal results in a drop of factor 3. If the analyst has a view on political developments, the PCA can guide toward the best expression: 1-factor residuals show the best country with which to express a general expectation about the euro crisis worsening or improving. And the PCA hedge ratios allow expressing a view on country-specific developments cleanly, isolating a trade on factor 2 or 3 (e.g. a view that Ireland will improve *relative* to other EUR sovereigns) from the overall level of the euro crisis (factor 1).

If the analyst has no political view and is looking for "pure" relative value trading opportunities, he should exclude the bailout countries from the PCA input data. This allows factors 2 and 3 to reveal the relative value mechanisms across the EUR sovereign CDS market, irrespective of the

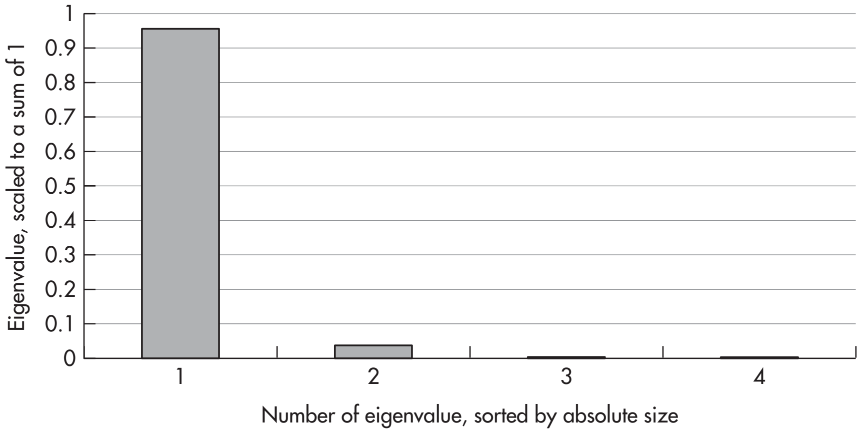

FIGURE 13.7 Scaled eigenvalues of a PCA on CDS quotes for core EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 6 May 2009 to 26 Sep 2012, weekly data.

political developments in a bailout country. Figure 13.7 and Figure 13.8 show the results of a PCA on the EUR sovereign CDS market, which has been restricted to the core countries Germany, France, the Netherlands, and Austria.

Like the PCA of the whole EUR sovereign CDS universe, the first factor represents the overall CDS level, producing a fever chart for the EUR crisis (Figure 13.9). Also, the relative sensitivity of individual countries as given by the first eigenvector is very similar to the first PCA. Unlike the first PCA, however, there are no country-specific factors anymore. Rather, factor 2 groups together Germany and France (negative sensitivities) and the Netherlands and Austria (positive sensitivities) and can thus be interpreted as "big versus small core country" factor. The scaled eigenvalues (Figure 13.7) suggest that the differentiation between big and small countries is in fact the only powerful mechanism (besides the overall CDS level represented by factor 1) in the core CDS market. And since the size of a country is independent of political influence, factor 2 of the core PCA can be considered a basis for "pure" relative value trades. This is further supported by the high speed of mean reversion of factor 2 (Figure 13.9). The differentiation between big and small core countries appears to be like statistical noise – unlike the differentiation of bailout versus non-bailout countries, which is subject to long-term (slow speed of mean reversion) political (little value of statistical models) decisions.

Thus, trades on factor 2 of the core PCA can justifiably be treated as 'pure' relative value positions, exploiting the statistical properties like mean reversion and being hedged (through PCA hedge ratios) against factor 1

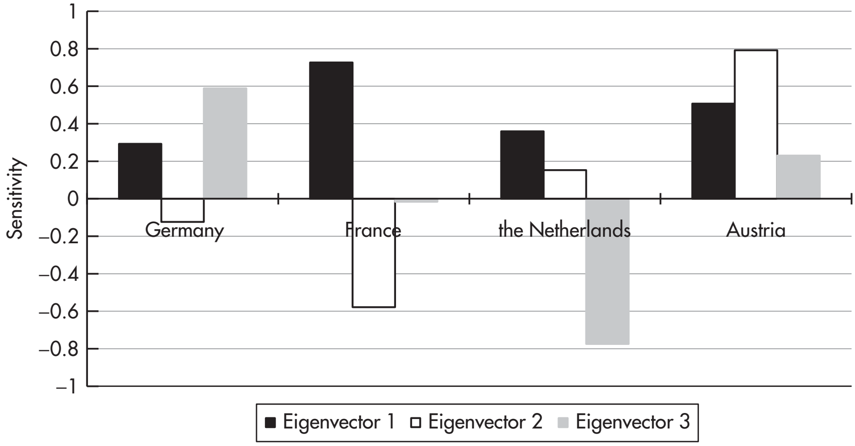

FIGURE 13.8 First three eigenvectors of a PCA on CDS quotes for core EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 6 May 2009 to 26 Sep 2012, weekly data.

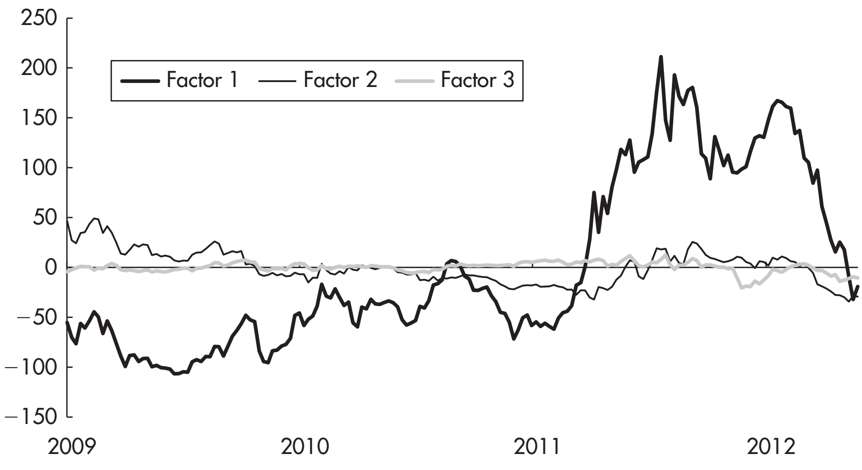

FIGURE 13.9 First three factors of a PCA on CDS quotes for core EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 6 May 2009 to 26 Sep 2012, weekly data.

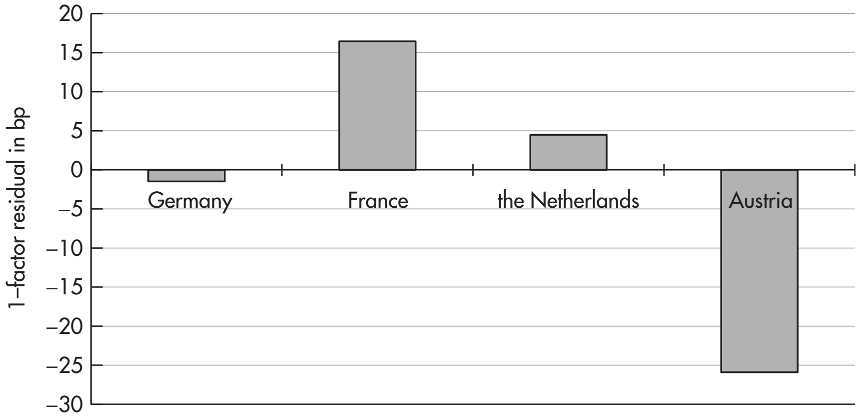

FIGURE 13.10 Current 1-factor residuals of a PCA on CDS quotes for core EUR sovereigns.
Sources: data – Bloomberg; chart – Authors.

Data period: 6 May 2009 to 26 Sep 2012, weekly data; "current" as of 26 Sep 2012.

(i.e. politically driven developments of the euro crisis and their impact on the CDS market). With factor 2 being currently considerably away from its mean, big core countries have CDS spreads that appear too high relative to small core countries. Figure 13.10 suggests that selling French versus buying Austrian CDS offers a 42 bp profit potential. The hedge ratio is given by the first eigenvector (buy 1.4 Austrian CDS for every French CDS sold) and should ensure that the performance of the trade is independent from political developments and their impact on the CDS market in general, which have caused the wild swings of factor 1 in Figure 13.9.

Of course, a risk to the position is that France drops out of the group of core countries and becomes a bailout country (with a country-specific factor like Ireland in the first PCA). However, given the high speed of mean reversion of factor 2, the trade is likely to perform before slow-moving macroeconomic events cause the market to reassess the status of France in the core group. Also, an investor concerned about France could still exploit the deviation of factor 2 from its mean by selling German or Dutch versus Austrian CDS, for an expected profit of 23 or 29 bp.

While a statistical analysis like the PCA above can provide interesting insights into the structure of CDS markets and lead to their exploitation through profitable relative value trades, we need to keep in mind that we are operating in an abstract world, assumed to be free from potential pitfalls. At the end of this chapter we shall describe ways to address these issues.

Other applications worth investigating along the same lines include the CDS quotes of a set of corporations of the same sector, or of the same country, or over different sectors, or over different countries.

Overall, we consider the CDS market as an excellent field to apply PCA, since:

- PCA can structure the information from a multitude of different bond issuers (different country, different rating, different sector) in the form of a few relevant factors, at the same time revealing significant interpretations.
- PCA thereby gives the basis for relative value trades within the credit universe, that is for exploiting mismatches between different issuers while being hedged against overall moves in the CDS market.

A PCA ON RISK-FREE BOND YIELDS

Combining CDS with their underlying bonds, one can create, analyze, and trade a proxy for risk-free government yield curves. This is another application of CDS which offers rather new analytical perspectives and RV trading strategies.

The theoretical benefit of this exercise is that we become able to analyze the factor structure of risk-free yield curves. In addition, we shall find that the statistical properties of risk-free yield curves are superior to those of unadjusted yield curves. In particular, problems like correlation between factors of a PCA during subperiods tend to be much less pronounced when using risk-free yield levels as input data.

On the practical side, analyzing the risk-free yield curve (i.e. the difference between the yield and CDS curve) gives insights into the pricing mechanisms of the bond yield curve *relative* to the CDS curve. These insights can then be translated into relative value positions on the yield versus CDS curve, for example, a steepening position on the JGB curve *relative* to the Japan CDS curve. What has been said above about applying PCA analysis to the CDS market is true for applications of PCA to risk-free bond yield curves as well: being among the first to gain insights into new types of relative value relationships, as in the shape of the bond yield curve *relative to* the CDS curve, suggests high returns from trading strategies based on these insights.

One has two different options to create synthetic risk-free bond yields by subtracting the CDS from the "normal" bond yield as traded in the market:

- Subtracting the adjusted CDS, i.e. without the DO and FX component as described above, gives a proxy for the risk-free yield curve. The bond yield can be considered a combination of the risk-free yield plus information about the market's assessment of its credit risk. Thus, subtracting the adjusted CDS quotes, i.e. only that part of the CDS quotes which actually represents credit exposure rather than optionality, from the bond yield can be considered an expression of the risk-free yield level. If the goal is only analytical, this option seems the natural choice. However, when also applied for trading, using the adjusted rather than the "full" CDS subjects the trading strategies to the significant risk of misestimation of the DO and FX component. Specifically in case of default, misestimation of the DO and FX components can lead to being underhedged. Imagine that in the situation of Figure 13.1 we estimated the JPY to weaken by 50% in case of default and therefore bought only 0.5 CDS per JGB (and assume a cash settlement). If default occurs and the JPY actually weakens by that amount, 0.5 CDS per bond are sufficient; but if the JPY weakens less than expected (maybe due to the large foreign reserves), we are underhedged.

- Subtracting the full (unadjusted, i.e. as observed in the market) CDS, i.e. the credit information together with the DO and FX component, results in an analytically inferior proxy for the risk-free bond yield, since it also contains information about the expected weakening of the FX rate of a defaulting country, for instance. On the other hand, given the nature of current CDS specifications, this is the only way to obtain trades which are sure to avoid the risk of being underhedged in case of default. We will use the definition "Risk-free yield = bond yield minus (unadjusted) CDS" in this section and end it by highlighting the pitfalls introduced by it.

Figure 13.11, Figure 13.12, and Figure 13.13 display the results of a PCA on the risk-free BTP (Italian government bond) yield curve in comparison with the PCA on the usual BTP yield curve. Note that the data input starts in 2006, that is it covers both a period when CDS were close to zero and exhibited virtually no volatility (see Figure 13.4) and a period when CDS quotes became a major driving force of bond yields. These results are typical and appear in a similar form in the US Treasury and JGB markets (and hence the outcome is not limited to Eurozone sovereigns), though the quantitative impact of the adjustment is sometimes less than in the case of Italy.

Starting the discussion of the results from the theoretical side, the data period covers the pre-crisis situation (2006, 2007), when credit considerations had a negligible impact on BTP yields. Actually, before the crisis, market participants usually considered these government bonds as risk-free and thus there was little difference between the two PCAs until 2008. (See Figure 13.12.) Starting in 2008, however, this assumption was revised and credit quality became a major driving force of government bond yields. Consequently, the two PCAs show a distinctly different behavior from 2008 onwards. (See Figure 13.12 and Figure 13.13.)

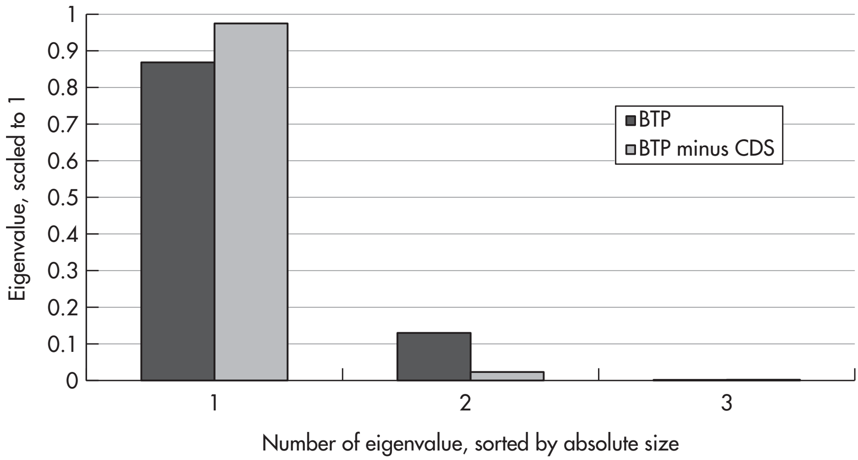

FIGURE 13.11 Scaled eigenvalues of a PCA on the risk-free BTP yield curve in comparison with those of a PCA on the BTP yield curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

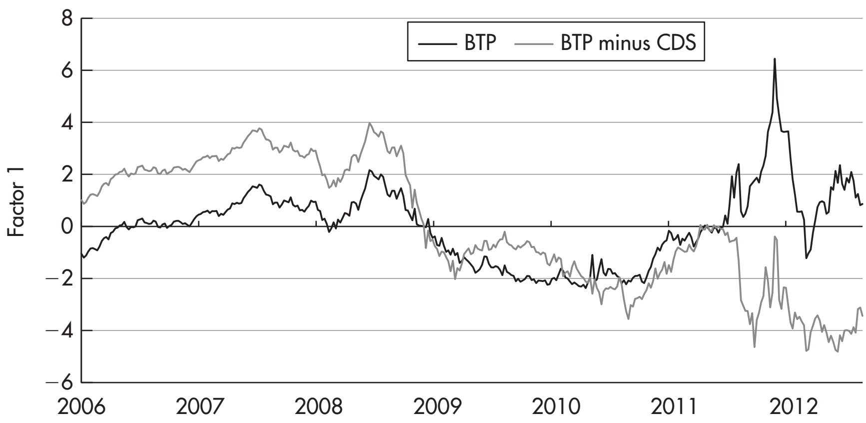

FIGURE 13.12 First factor of a PCA on the risk-free BTP yield curve in comparison with the first factor of a PCA on the BTP yield curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

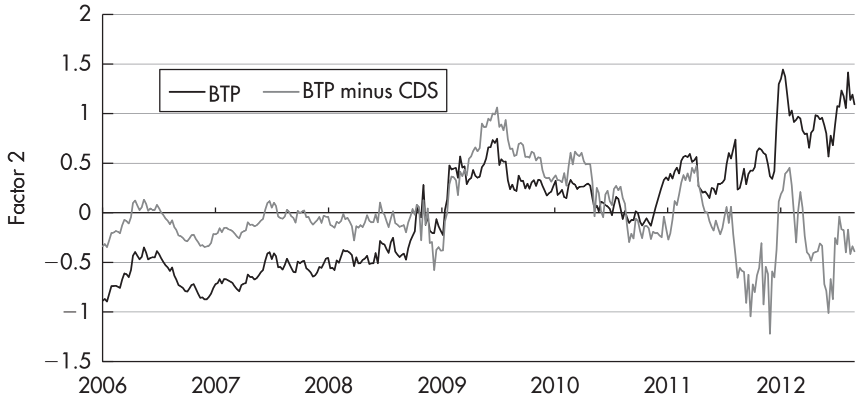

FIGURE 13.13 Second factor of a PCA on the risk-free BTP yield curve in comparison with the second factor of a PCA on the BTP yield curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

Moreover, the PCA of risk-free bond yields always uses default-free data, both before and after the credit crisis started. On the other hand, the benefit of excluding impacts from credit on the bond yield curve by subtracting CDS levels could introduce statistical noise if the CDS quotes were erratic. By contrast, the PCA of normal bond yields covers both a period when government bonds were considered default-free and a period when credit concerns drove most of the price action along the bond yield curve.

This caused a break in the statistical properties of the bond yield curve when the credit crisis started, which is visible in a number of statistical problems.

- Perhaps most important, the emerging credit concerns in 2008 had an impact both on the level (factor 1) and on the non-directional steepness (factor 2) of bond yield curves, since credit quality affects longer maturities more than it affects shorter ones.[10] This is the main reason behind the problem of correlation between factors 1 and 2 during subperiods

[10]This statement can be backed up with ratings transition matrices, which show that the yearly default probability increases with the length of exposure – unless the credit quality is very bad at the beginning, which was not the case for the sovereign examples used here.

of PCAs spanning pre-crisis and post-crisis data. We have discussed in Chapter 3 that this is a major pitfall for PCA-based trades. Thus, we are excited to find that by using risk-free bond yields as input variables we can exclude by construction the impact of credit on every factor of the PCA and thereby also get rid of a major source of problematic correlation between PCA factors in subperiods.[11] Proving this important point, we display in Figure 13.14 the correlation between factors 1 and 2 in the subperiod since 2010 of a PCA using normal BTP yields and of a PCA using risk-free BTP yields since 2006 as input. We observe that the problematic correlation between factors 1 and 2 of a PCA using normal BTP yields (Figure 13.14, top graph; compare also with Figure 3.23) is largely absent in a PCA using risk-free BTP yields (Figure 13.14, bottom graph).

- Thus, a shift in credit regime causes a correlation between factor 1 and factor 2 in a PCA of normal bond data (in the subperiod following the shift in credit regime). This results in factor 2 being driven in large measure by the directional factor 1 following a change in credit assessment. Intuitively speaking, part of the explanatory power of factor 1 shifts to factor 2, destroying its interpretation as non-directional steepness and the performance of any relative value trades based on it.
- Correspondingly, the explanatory power of the first factor decreases significantly. (See Figure 13.11.) Using risk-free yield levels, however, maintains a high explanatory power of the first factor. In fact, the results of a PCA on risk-free yield levels are very similar to those of a PCA of normal bond yields, which uses only data pre-crisis or only data post-crisis. (Compare the charts above with those in Chapter 3, in particular, Figure 13.11 with Figure 3.4.) In a manner of speaking, subtracting CDS quotes from bond yields maintains the usual three-factor structure of bond yield curves *independent of credit impacts*. While we have seen above that the credit component does nothing to *explain* the three-factor structure of bond yield curves, accounting for it can *maintain* the normal mechanisms of yield curves also in times of major shifts in the assessment of the credit quality of government bonds. The first factor loading of the CDS curve actually seems to have precisely the shape needed to correct the three factor loadings of the bond curve, so that they remain stable in changing credit regimes.

This is a crucial condition to conduct analysis in times of shifting credit assessments – and to construct trades which are hedged against their impacts on the bond yield curve. Imagine we had put on a 2Y-10Y

[11]Of course, there can still be a correlation between factors in subperiods for other reasons and we advise to check for it also when running a PCA on risk-free yields.